

Project Proposal

Production Label Printing System (PLP) with QR Code — Phase 1

Poly Packaging Services | Prepared: May 2026 | Version 1.0

1. Project Overview

A cloud-hosted, mobile-accessible web application is required to manage and print production LOT labels with QR codes across five production sections: Mixing, Extruder, Printing, Gusset, and Cutting. Manual label generation will be eliminated, traceability will be improved, and management dashboards and reports will be provided.

Key objectives:

- Auto LOT number generation with QR code labels (JobProfile embedded)
- Full traceability across all production sections
- Role-based access with audit trail and email alerts

2. Phase 1 Scope

CATEGORY	DELIVERABLES
User & Access	User Management, Feature-based RBAC, Audit Trail Logs, Auto Email Alerts
Dashboards	Directors Dashboard, Managers Dashboard
Masters (7)	Job Profile, Customer, Machine/Section, Supervisor & Operator, Work Shift, Material Type, Product Sizes Grouping
Label Printing (5)	Mixing, Extruder, Printing, Gusset, Cutting — each with Double Sticker (Paper Core + Film), QR Code (JobProfile), Transfer Note
Reports	5 Basic Reports with Export to Excel

3. Proposed Technology Stack

LAYER	TECHNOLOGY	NOTES
Frontend	React (PWA)	Mobile-accessible, installable on factory floor devices
Backend	Node.js	REST API,
Database	PostgreSQL	Cloud-hosted, no licensing cost
Hosting	Contabo VPS	~USD 20/mo — sufficient for 5 concurrent users
Security / WAF	Cloudflare (Free)	DDoS protection, HTTPS, DNS management, Reverse Proxy

4. Estimation & Investment

Development Timeline	~50 days
Dev Cost (mid-level)	~LKR 440,000
Cloud Infrastructure	~ USD 20/mo